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Kerala Digital Out-Of-Home Network

End-to-end execution blueprint for a six-node pilot across Kochi, Kozhikode and Kannur — compliance, sourcing, leases, finance, coastal hardening and automation.

GEOGRAPHY

Kochi · Kozhikode · Kannur

NODES

2 dual-face outdoor unipoles + 4 indoor MUPI kiosks

OUTDOOR SPEC

P3.91 / P4 · IP65/66 · ≥ 5,500 nits · C5-M

INDOOR SPEC

Dual 55-inch 2,500–3,000 nit · 8 mm glass

TARGET PAYBACK

15.9 months on the volume OEM stack

DOCUMENT DATE

07 September 2026

Prepared as an operations and investment working paper. Advertisement tax, customs classification, stamp duty and KSEB tariff must be confirmed against the live gazette before money moves. Courts: Ernakulam / Kozhikode / Kannur.

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0. Executive summary

DECISION SNAPSHOT

Build dual-face outdoor unipoles on private land outside NH 66 RoW. Lease mall kiosks on a hybrid floor-plus-18% model. Import or CKD only after factory salt-spray and 72-hour burn-in. Cast outdoor foundations before 1 June. Use the target hardware/sales stack if a 12–16 month payback is a board constraint.

BASE

CAPEX **₹1.95 Cr**	REVENUE / MO **₹16.03 L**	EBITDA / MO **₹6.74 L**
PAYBACK **29.0 mo**	MARGIN **42.0%**	ANNUAL ROI **41.4%**

TARGET

CAPEX **₹1.54 Cr**	REVENUE / MO **₹18.57 L**	EBITDA / MO **₹9.69 L**
PAYBACK **15.9 mo**	MARGIN **52.2%**	ANNUAL ROI **75.7%**

This report is the execution pack for a **six-node Digital Out-Of-Home (DOOH) pilot** across Kerala's three strongest north-and-central media markets. The network is two dual-face outdoor LED unipoles (Edappally / Kakkanad and Thondayad Bypass) plus four indoor dual-sided 55-inch MUPI kiosks (Lulu Mall, Forum Mall, HiLite Mall, Secura Centre / Capitol Mall).

Hardware envelope. Outdoor: P3.91 or P4 SMD LED, die-cast aluminium, IP65/IP66, ≥ 5,500 nits auto-dimming, silicone conformal coat, coastal C5-M paint. Indoor: dual 55-inch 2,500–3,000 nit commercial 4K behind 8 mm toughened glass. Players: NovaStar Taurus TB60 / TB30 with PiSignage or VNNOX, offline cache, and Hivestack / Lemma programmatic tags.

The P&L hinge. The stated slot card (Kochi outdoor ₹50,000 / Kozhikode outdoor ₹30,000 / mall ₹8,500 per slot per month) **does not repay a flagship China import in 12–16 months**. Dual-face outdoor inventory, hybrid leases (minimum guarantee vs 18% of site gross), and a volume OEM FOB are what land payback inside the target band.

Scenario	Occupancy	LED FOB	Economic CAPEX	EBITDA / month	Payback
Base (premium import)	75%	USD 650 / sq.m	₹1.95 Cr	₹6.74 L	29.0 months
Target (volume CKD)	90%	USD 330 / sq.m	₹1.54 Cr	₹9.69 L	15.9 months
Conservative HS 85285900	60%	USD 650 / sq.m	₹2.05 Cr	₹4.36 L	47.0 months

Live numbers are produced by `dooh_kerala_calculator.py`. Statutory rupee rates (advertisement tax, stamp duty, KSERC slabs, BCD) must be confirmed with the corporation, KSERC, CBIC and a Kerala-enrolled advocate before capital is committed.

Network topology and design envelope

Pilot topology: Kochi (1 dual-face outdoor unipole at Edappally/Kakkanad + 2 indoor MUPI kiosks at Lulu Mall and Forum Mall) · Kozhikode (1 dual-face outdoor unipole at Thondayad Bypass + 1 indoor MUPI at HiLite Mall) · Kannur (1 indoor MUPI at Secura Centre / Capitol Mall).

Display spec: Outdoor P3.91 (or P4) SMD LED, die-cast aluminium, IP65/IP66, ≥ 5,500 nits with auto-dimming, conformal-coated PCBs. Indoor: dual-sided 55-inch 2,500–3,000 nit commercial 4K, 8 mm toughened glass kiosk.

Software: NovaStar Taurus TB30/TB60 + cloud CMS (PiSignage or VNNOX) + offline cache + Hivestack / Lemma programmatic tags.

Figures in Module 4 are produced by `dooh_kerala_calculator.py`. Statutory amounts are **working planning rates** as of 2026; confirm the live gazette / council schedule before paying.

Module 1 — Administrative, regulatory and compliance roadmap

1.1 Legal spine

Instrument	What it controls	Portal / office
Kerala Municipality Act, 1994, ss. 271–272	Advertisement tax + written permission of the Secretary before any display	Corporation Town Planning / Revenue
Kerala Municipality Building Rules, 2019	Unipole treated as a building / advertising sign; structural drawings	IBPMS: https://buildingpermit.lsgkerala.gov.in
Kerala Industrial Single Window Clearance Boards Act (K-SWIFT)	Parallel departmental clearances for the enterprise (KSEB, Electrical Inspectorate, Fire, Municipality)	https://kswift.kerala.gov.in
Control of National Highways (Land and Traffic) Act, 2002 + IRC: 46-1972	No commercial hoarding **inside NH RoW**; setbacks beyond RoW	NHAI Project Director (NH 66) / PWD Roads
Electricity Act 2003 + Kerala Electricity Supply Code 2014	LT connection, metering, duty	https://wss.kseb.in
CEA Safety Regulations + Kerala Electrical Inspectorate	Scheme approval / sanction for energisation of advertising installations	https://ceisuraksha.ceikerala.gov.in

Kochi Corporation's 2012 advertisement bye-law still matters operationally: Secretary permission is mandatory on public **and** private land; hoardings generally **≥ 1 m** from thoroughfares; LED boards have historically been restricted from screening films (still / slide-style creatives were the bye-law default). In 2025 the Corporation began a designated-site LED programme and the Town Planning Committee is reviewing digital boards that encroach footpaths. **Do not install on footpath, median, or carriageway. Prefer private plots outside RoW, or Corporation-tendered designated sites.**

1.2 K-SWIFT + municipal application (step by step)

Step 0 — Constitute the applicant. Private limited company with PAN, GSTIN (Kerala registration), IEC (for imports), and a local authorised signatory. Open a current account with an AD code for ICEGATE.

Step 1 — K-SWIFT registration. Create an individual login, then an enterprise profile, at https://kswift.kerala.gov.in. Run the "identify clearances" wizard. Tick Municipality (building permit + trade/advertisement), KSEB (new LT), Electrical Inspectorate, and Fire & Rescue (mall kiosks). K-SWIFT is a router — it does **not** replace the Secretary's s. 272 permission.

Step 2 — Building permit (IBPMS). File through https://buildingpermit.lsgkerala.gov.in (also linked from Kochi Corporation: http://kochicorporation.lsgkerala.gov.in). Upload:

1. Site plan (1:200) with NH/SH edge, RoW, setbacks, existing trees, KSEB lines.
2. Structural drawings + certificate (see `docs/legal/STRUCTURAL_STABILITY_CERTIFICATE.md`).
3. Soil report (two boreholes) for outdoor sites.
4. Proof of title / consent (sale deed, tax receipt, possessor's NOC).

5. AAI NOCAS print for Kochi if the tip of the unipole is in CIAL obstacle-limitation surfaces: https://nocas2.aai.aero/nocas/
6. Lightning and earthing schematic.

Step 3 — Advertisement permission (s. 272). Separate application to the Corporation Secretary (Town Planning Standing Committee file). Attach the building permit (or acknowledgement), tax challan, RFID/identification as required by bye-law, and a content policy (no tobacco, no film screening if still banned, auto-dimming affidavit).

City portals:

City	Civic body	Working entry points
Kochi	Kochi Municipal Corporation	http://kochicorporation.lsgkerala.gov.in · Citizen service / IBPMS
Kozhikode	Kozhikode Municipal Corporation	https://kozhikodecorporation.lsgkerala.gov.in
Kannur	Kannur Municipal Corporation	https://kannurcorporation.lsgkerala.gov.in

Step 4 — Pay advertisement tax before the permission is signed. The Secretary cannot grant s. 272 permission if tax is unpaid (s. 272(2)(ii)).

1.3 Advertisement tax — formula (Class A vs Class B)

The Kerala Municipality Act does **not** freeze a statewide rupee rate. Each Council notifies a schedule with Government approval. Until you hold the current resolution, budget with the formula below and treat the rates as a **planning envelope** (verify at the Revenue counter; they change by council resolution).

```
Annual tax (₹) = Display area (sq.m) × Zone rate (₹/sq.m/year) × Type factor
```

Zone	Typical roads in this pilot	Planning rate (static illuminated)	Type factor for digital / LED
Class A	Kochi Edappally bypass, Kakkanad InfoPark corridor, MG Road / Banerji Road class roads	₹2,500 / sq.m / year	**2.0** → ₹5,000 / sq.m / year
Class B	Kozhikode Thondayad, Kannur mall precincts, secondary arterials	₹1,500 / sq.m / year	**2.0** → ₹3,000 / sq.m / year

Worked outdoor Kochi dual-face: 32 sq.m × 2 faces × ₹5,000 = **₹3,20,000 / year** (₹26,667 / month).
Worked outdoor Kozhikode dual-face: 32 × 2 × ₹3,000 = **₹1,92,000 / year**.
Indoor MUPI (two 55-inch faces ≈ 1.05 sq.m each, treated as illuminated indoor): budget ₹14,400–₹24,000 / year per kiosk.

Illuminated / digital multipliers of 1.5×–2.5× are the industry pattern in Kerala municipal schedules. If the live bye-law prices digital as a lump-sum licence instead of per sq.m, pay the lump sum and keep the per-sq.m figure as a sensitivity in the calculator (`ad_tax_annual_inr` on each `SiteSpec`).

1.4 PWD and NHAI (NH 66, state highways, Thondayad, Edappally)

Inside National Highway RoW: commercial hoardings are prohibited. MoRTH circulars (including 16 May 2002 and the 2016 reminder) and the Control of National Highways (Land and Traffic) Rules, 2002, treat advertisement structures on NH land as encroachments. Kerala High Court directions (including *D.B. Binu v. State of Kerala* on the Edappally–Vyttila stretch) require Secretaries and NHAI Project Directors to remove illegal boards.

Siting rule for this pilot: take a **private plot outside the notified RoW**, then apply IRC:46-1972 / NBC signage rules:

Rule	Number
Minimum setback from edge of carriageway	**10 m** (IRC:46). Kerala outdoor-ad policy submissions to the High Court have also used **50 m from road/footpath edge** for certain categories — design to the **stricter** of IRC 10 m and the local PWD/NATPAC condition on the NOC.
Distance from junction, bridge, or railway crossing	**100 m** (50 m in built-up urban sections if PWD agrees in writing).
Distance from an official traffic sign	**50 m** along the road.
Area vs setback (IRC)	Advertisement area ≤ **0.3 sq.m per metre of setback**. A 64 sq.m dual-face equivalent needs a generous setback or an explicit PWD/NHAI relaxation; this is why designated Corporation sites or deep private plots matter.
Median / footpath / green strip	**Zero**. Do not bid these.

NH 66 NOC workflow

1. Identify the chainage on the latest NHAI alignment (Edappally / Koonammavu side for Kochi; Ramanattukara–Thondayad for Kozhikode if the plot touches NH).
2. If the plot is **outside RoW**, apply to the **NHAI Project Director, PIU Kozhikode / PIU Kochi** for a no-objection on traffic-safety grounds, with IRC:46 checklist, sight-line drawing, and brightness affidavit. Use NHAI LMS / PIU email as directed by the PIU.
3. If any cable, crane swing, or stay enters RoW, stop — that is an encroachment application and will fail.
4. For **State Highway / PWD** roads (Thondayad Bypass sections not handed to NHAI): apply to the Executive Engineer, PWD Roads, with the same pack plus District Road Safety Council NOC.

1.5 KMBR, wind 39 m/s (140 km/h), soil, lightning

IS 875 Part 3 basic wind speed for Kochi, Kozhikode, Thiruvananthapuram, and Kannur region: **Vb = 39 m/s ≈ 140 km/h**. Design example is in the structural certificate template.

Mandatory outdoor pack:

1. Structural design in STAAD/ETABS, steel per IS 800, foundation per IS 456.
2. Geotech: two boreholes, SPT, water table. Coastal / canal sites (Edappally) → raft or piles, not isolated footings in high water table.
3. Importance factor k1 ≥ 1.06; seismic Zone III (IS 1893).
4. Lightning protection IS/IEC 62305 Level III; earth **≤ 2 Ω**.
5. Annual monsoon recertification by 15 May.

1.6 KSEB power — LT-VII(A), not LT-VII(C)

LT-VII(C) is for cinema, multiplex, auditorium, stadium. A roadside LED plant is **LT-VII(A) Commercial**. Indoor mall taps ride the mall's commercial connection via sub-meter.

KSERC tariff used in the calculator (Gazette order w.e.f. 05 December 2024, three-phase LT-VII(A)):

Component	Rate
Fixed charge	**₹190 / kW / month** (three-phase)
Energy (non-telescopic)	0–100 u: ₹6.05 · 0–200: ₹6.80 · 0–300: ₹7.50 · 0–500: ₹8.15 · **above 500: ₹9.40 / kWh**
Electricity duty	**10% of energy charges** (Kerala Electricity Duty Act, 1963)
Planning fuel / FPPCA buffer	₹0.50 / kWh

Outdoor dual-face at 24 kW connected, 65% load factor, 18 h/day lands in the top slab (~8,400 kWh/month) → about **₹96,000 / month** per unipole including duty (see calculator). Sanction **25 kW** dual-face / **15 kW** single-face.

Application checklist (https://wss.kseb.in)

1. Online LT new connection, ₹50 application fee.
2. Proof of identity, ownership/consent, GSTIN.
3. Sanctioned load calculation: LED 600–800 W/sq.m peak × area × faces × 0.8 diversity + controller 0.5 kW + lighting 0.2 kW.
4. Test report from a licensed electrical contractor; ELCB, SPD, earthing drawing.
5. Security deposit and service-line estimate per KSERC cost data.
6. Electrical Inspectorate: advertising / neon-class installations and connected load in this range go through https://ceisuraksha.ceikerala.gov.in for scheme approval and sanction for energisation (Kannur Inspectorate explicitly lists neon sign boards).
7. Meter: three-phase static meter, **ToD-capable**, so that if KSEB extends ToD to this LT commercial node you do not replace hardware. Dual-tariff / ToD is not universally applied to LT-VII(A) today; specify it in the application as a future-proof requirement.

Mall indoor: 0.55 kW, 14 h, 20% mall markup → about **₹2,100 / month** per kiosk.

Module 2 — Hardware sourcing and procurement

2.A China OEM (Shenzhen cluster)

OEM	URL	Why they are on this shortlist
Unilumin	https://www.unilumin.com	Ushin / Usurface outdoor, IP66, marine coating options
Absen	https://www.absen.com	A-series outdoor advertising, strong India service footprint
Leyard / Planar	https://www.leyard.com	Custom high-brightness outdoor
LianTronics	https://www.liantronics.com	High-reliability outdoor cabinets
Apexls	http://www.apexls.com	Custom kiosks and outdoor panels; typically keener FOB

Commercial posture: `base` scenario prices **USD 650 / sq.m FOB** (flagship IP66, 5,500 nits, spare kit). `target` scenario prices **USD 330 / sq.m FOB** (volume A-series / Apexls CKD with the same IP and coating spec — only award if FAT in `docs/procurement/QC_AND_IMPORT_CHECKLIST.md` passes). Do not buy “P3.91 outdoor” without salt-spray data.

HS codes and duty (planning)

Complete advertising video walls are **Heading 8528**. Prefer **CTH 8528 52 00** (monitors capable of connecting to an ADP machine / player) where CESTAT / CAAR support exists for LED monitor tiles. Customs may reclass to **8528 59 00** (“other monitors”, **20% BCD**). Do not use 8531 (indicator panels) for video walls.

Duty stack on 8528 52 00 used in the calculator:

```
BCD  = 10% of CIF
AIDC = 10% of BCD
SWS  = 10% of (BCD + AIDC)
IGST = 18% of (CIF + BCD + AIDC + SWS)
Economic cost (ITC of IGST claimed) = BCD + AIDC + SWS = 12.1% of CIF
Cash at port                        = 12.1% + IGST ≈ 32.3% of CIF
```

Run `--conservative-hs` to force 20% BCD. File an advance ruling (CAAR Mumbai) if the first container is > USD 80,000.

Logistics: FCL 40’ Shekou / Yantian / Nansha → **INCOK, ICTT Vallarpadam** (DP World). CHA files BE on https://www.icegate.gov.in. Port: https://www.cochinport.gov.in · terminal: https://www.dpworld.com/en/ports-terminals/india/ictt/services. Do not destuff in rain.

2.B India stockists and kiosk fabricators

Vendor	URL	Use
Vishwanjali Technology	https://www.vishwanjali.com	Cabinet assembly, indoor/outdoor LED
Elpro Technologies	https://www.elprotech.in	Standees and indoor/outdoor kiosks
Matrix Digital Corporation	https://www.matrixdisplaysystem.com	Modules, PSU, die-cast cabinets

Vendor	URL	Use
Pixel LED India	https://www.pixelledindia.com	Receiving cards, NovaStar, emergency modules
Mean Well India	https://www.meanwell.in	HLG/ELG weatherproof PSU

Split buy: import LED modules + receiving cards; fabricate unipoles and MUPI shells in Kochi (IS 2062 steel, Akzo/Jotun C5-M paint system); buy Mean Well locally for 7-year replacement. Hold a spare kit in a **dry Kochi warehouse**, never on the unipole walkway.

Controllers: NovaStar Taurus **TB60** outdoor, **TB30** indoor. CMS: https://pisignage.com or VNNOX https://www.vnnox.com. Programmatic: https://www.hivestack.com · https://lemmatechnologies.com.

Module 3 — Site leasing and mall pitch

3.1 Pitch deck for HiLite, Lulu, Forum, Gokulam, Secura (12 slides)

1. **Cover** — Kerala coastal DOOH, six-node pilot, Kochi HQ.
2. **Why this mall** — catchment (Lulu: regional + tourist; HiLite: Malabar affluent; Secura/Capitol: Kannur city).
3. **Product** — dual-face 55” 3,000 nit, 8 mm glass, 1.2 × 1.2 m footprint, silent, no audio by default.
4. **Look & feel** — Spectrum of night/day renders in the actual atrium (commission a local 3D artist; do not pitch generic Shenzhen photos).
5. **Mall inventory** — 12% reserved loop for mall events (Clause 8 of the MoU).
6. **Safety** — ELCB, SPD, toughened glass, fire-load letter, night install window.
7. **Power** — sub-meter, 20% cap on markup, no phantom HVAC lump sum.
8. **Commercial** — hybrid: Minimum Guarantee + 18% of kiosk gross, whichever higher.
9. **Content standards** — no tobacco, brightness caps, proof-of-play portal login for the mall GM.
10. **Insurance** — ₹2 Cr public liability, mall as additional insured.
11. **Case references** — Indian mall DOOH comps (keep honest; Kochi digital is still young).
12. **Ask** — 36-month licence, 18-month lock-in, 3-month deposit, LOI in 21 days.

Outdoor landowner pitch is shorter: private plot, outside RoW, 18% or hybrid with ₹80,000 / ₹45,000 floor, monsoon indemnity, crane access four times a year.

Full clause set: `docs/legal/SITE_LEASE_AND_MOU.md`.

3.2 Deal physics (why hybrid)

A Lulu kiosk at ₹8,500 × 12 slots × 75% occupancy yields ~₹81,000 / month. A ₹75,000 fixed rent leaves almost nothing after power and CMS. Hybrid floors (₹20,000–₹40,000 indoor, ₹45,000–₹80,000 outdoor) plus 18% of gross keep the mall/owner whole in a good month without killing the pilot. The calculator’s `--lease-model hybrid|fixed|revshare` switches this live.

Module 4 — Financial model and ROI

Inventory assumption: **10-second spots in a 100-second loop** = 10 slots per outdoor face × 2 faces = 20 outdoor slots; indoor 6 slots × 2 faces = 12 slots.

Rate card (as specified):

Inventory	₹ / slot / month
Kochi outdoor	50,000
Kozhikode outdoor	30,000
Mall MUPI (all three cities)	8,500

Theoretical 100% direct capacity = **₹20.08 lakh / month**.

4.1 CAPEX — `base` (premium import, USD 650/sq.m, dual-face)

Economic CAPEX **₹1.95 Cr**. IGST cash at port **₹15.1 lakh** (ITC). Cash build **₹2.10 Cr**.

Line	₹
Outdoor LED FOB (4 faces × 32 sq.m)	69,88,800
Freight + insurance + ICTT	4,80,000
BCD + AIDC + SWS	9,03,725
Unipole, foundation, soil, PE	40,94,000
Electrical, SPD, lightning	7,56,000
KSEB 3-ph connection	5,50,000
NovaStar + player + 4G	4,62,000
K-SWIFT / municipal / PWD year-1	5,00,000
Outdoor install + 72 h burn-in	4,50,000
Indoor 55” pairs (4 kiosks)	8,80,000
MUPI fabrication	7,40,000
Indoor player + install	4,00,000
CMS + Lemma/Hivestack hook	2,50,000
Spares warehouse	6,30,000
Contingency 8%	14,46,762
Economic total	**1,95,31,287**

4.2 OPEX and revenue — `base` (75% occupancy, 35% programmatic fill of unsold at 55% of direct CPM)

	₹ / month
Fixed (hybrid leases, KSEB, tax, CMS, 4.5% AMC, ops ₹60k, insurance ₹15k)	7,68,434
Sales commission 10%	1,60,264
OPEX	**9,28,698**
Direct sold	15,06,000
Programmatic	96,635
Gross revenue	**16,02,635**
Kochi / Kozhikode / Kannur	9.61 L / 5.60 L / 0.81 L
EBITDA	**6,73,937**
Net operating margin	**42.1%**
Simple payback	**29.0 months**
Annual ROI on economic CAPEX	**41.4%**
Direct occupancy breakeven	**22%**

Outdoor KSEB is the second-largest site cost after lease (~₹96,000 / unipole / month). Auto-dimming is a P&L control, not just a safety feature.

4.3 `target` stack — 12–16 month payback (15.9 months in the calculator)

The specified rate card **cannot** repay a USD 650/sq.m dual-face import in 16 months without occupancy and hardware changes. The `target` scenario is the execution path that does:

1. Volume OEM / CKD FOB **USD 330 / sq.m** with the same FAT (not a spec downgrade on IP or nits).
2. **90%** direct occupancy (two national retainers + local jewellery / education / healthcare / QSRs pre-sold before energisation).
3. In-house sales, commission **6%**.
4. AMC **3.5%** (own technician + Pixel LED / Mean Well India spares).

Result: economic CAPEX **₹1.54 Cr**, gross revenue **₹18.57 lakh / month**, EBITDA **₹9.69 lakh / month**, margin **52.2%**, payback **15.9 months**, annual ROI **75.7%**.

4.4 Programmatic

Unsold loop time is offered to Hivestack / Lemma at **55% of the direct slot CPM**. Fill 35% (`base`) or 45% (`target`) of unsold. Do not cannibalise direct by opening programmatic until occupancy is below 85% or the slot is inside 72 hours of play. Proof-of-play logs are the audit trail for both mall revenue-share and SSPs.

GST 18% on advertising (SAC 9983) is treated as a pass-through and is **not** inside EBITDA.

Module 5 — Technical deployment and weather hardening

Kerala's design enemies are **salt (Kochi backwaters, Kozhikode coast), 3,000 mm rain, 90% RH, and 39 m/s gusts**, not desert heat.

5.1 Cabinets and coating

Choice	Spec
Cabinet	Die-cast aluminium, not iron. Iron rusts through a monsoon even with powder coat.
Gasket	EPDM, continuous, no splices at corners.
Fasteners	A4 / 316 stainless, anti-seize on threads.
Paint	ISO 12944 C5-M, zinc-rich primer + epoxy + PU, DFT ≥ 240 µm on the unipole.
PCB coat	**Silicone (SR)** IPC-CC-830, not acrylic. Acrylic micro-cracks in salt air.
Modules	Rear-service, IP65, conformal-coated driver ICs, potting on power plugs.
Drains	Two weep holes per cabinet, insect mesh, never silicone-sealed shut.

Indoor kiosks: 8 mm toughened glass, powder-coated aluminium frame, hidden 120 mm fans with G4 filters, condensate path away from PSUs. Mall AC is not a substitute for a sealed player enclosure.

5.2 Electrical safety

1. TT earthing, two pits minimum, **≤ 2 Ω**, 3 m apart, inspected every April and September.
2. **Class II SPD** (Type 2, 40 kA 8/20) at the LT incomer; Type 3 at the player.
3. **ELCB / RCCB 100 mA** incoming, 30 mA on socket/player circuit.
4. Lightning IS/IEC 62305 Level III, down-conductor not shared with data screen.
5. Mean Well HLG, derate 20% below nameplate in 40 °C cabinet.
6. No outdoor transformers in Corporation limits (Kerala Electrical Inspectorate guidelines). Stay on KSEB LT.

5.3 Controllers, CMS, failover

```
Brand MP4 ──► PiSignage / VNNOX cloud ──► HTTPS ──► TB60 (outdoor) / TB30 (indoor)
                                                   │
                                                   ├─ 4G primary (Jio) + 4G failover (Airtel)
                                                   ├─ local SSD: last-known 100-second loop (6
creatives)
                                                   └─ watchdog: if cloud > 90 s unreachable →
play /offline/loop/
```

Failover logic (implement in the player agent):

1. Heartbeat to CMS every 30 s.
2. If 3 consecutive misses, freeze the current playlist from local cache; do **not** show a black frame or an OS desktop.

3. If cache is empty, play a licensed public-service still (monsoon helpline / 112) — never a competitor or political creative.
4. When the link returns, do not interrupt a 10-second spot; splice at the next loop boundary.
5. Proof-of-play continues locally (CSV + hash) and uploads in catch-up; this is what Lemma/Hivestack and mall auditors will demand.

Brightness: light sensor on the **north** face, 5,500 nits cap day, 600 nits after 22:00, 15% per minute ramp. Log nits with proof-of-play.

NovaStar: https://www.novastar.tech · PiSignage: https://pisignage.com · VNNOX: https://www.vnnox.com.

Module 6 — Automated execution script and mobilisation

The calculator is `dooh_kerala_calculator.py` at the repository root (Python 3.10+, stdlib only).

```
python3 dooh_kerala_calculator.py --scenario base
python3 dooh_kerala_calculator.py --scenario target
python3 dooh_kerala_calculator.py --scenario conservative --conservative-hs
python3 dooh_kerala_calculator.py --occupancy 0.8 --lease-model revshare --outdoor-faces 1
python3 dooh_kerala_calculator.py --json --out /tmp/dooh.json
python3 -m unittest tests.test_dooh_kerala_calculator -v
```

Flag	Effect
`--scenario base|target|optimistic|conservative`	Occupancy, programmatic fill, FOB, commission, AMC
`--occupancy 0–1`	Direct sold share of loop
`--programmatic-fill 0–1`	Share of *unsold* inventory sent to SSPs
`--kochi-slot` `--kozhikode-slot` `--mall-slot`	Rate card
`--lease-model hybrid|fixed|revshare`	18% / floors
`--outdoor-faces 1|2`	Single vs dual unipole
`--fob-usd`	LED FOB
`--commission`	Sales commission
`--usd-inr`	FX
`--conservative-hs`	20% BCD under 8528 59 00

Formulas (same as the script):

```
kWh         = kW_connected × load_factor × hours/day × 30
energy      = kWh × LT-VII(A) non-telescopic slab rate
power_bill  = energy + (190 × sanction_kW) + 0.10×energy + 0.50×kWh
site_revenue = occupancy×slots×price + (1−occupancy)×slots×price×prog_fill×0.55
lease       = max(minimum_guarantee, 0.18 × site_revenue)   # hybrid
EBITDA      = Σ site_revenue − leases − power − tax − CMS − AMC − ops − insurance −
commission
payback     = economic_CAPEX / monthly_EBITDA
ROI_annual  = 12 × EBITDA / economic_CAPEX
```

90-day mobilisation (outdoor first)

Week	Action
1–2	Incorporate, GST, IEC, K-SWIFT login, shortlist two Kochi plots **outside NH RoW**, issue mall LOIs

Week	Action
3–4	Geotech + topographic survey + IRC setback drawing; file IBPMS + s. 272; start KSEB WSS
5–6	FAT at OEM or India assembler; book 40’ to Vallarpadam; Structural Engineer certificate
7–8	Inspectorate scheme; unipole fabrication in Kochi; mall kiosk mock-up sign-off
9–10	Foundation + earth pits before monsoon (do not open foundations after 1 June)
11–12	Cabinet hang, 72 h burn-in, CMS + programmatic tag, first paid loop

If the calendar hits June before foundations are cast, **slip outdoor to September** and go live on the four indoor kiosks first. A half-built unipole in a Kerala monsoon is a High Court case waiting to happen.

Document map

File	Role
`docs/legal/SITE_LEASE_AND_MOU.md`	Full clause set (hybrid rent, KSEB, 12% mall loop, monsoon indemnity)
`docs/legal/STRUCTURAL_STABILITY_CERTIFICATE.md`	IS 875 39 m/s sign-off
`docs/procurement/QC_AND_IMPORT_CHECKLIST.md`	FAT, salt spray, ICEGATE
`dooh_kerala_calculator.py`	Live P&L

Rates cited: KSERC LT-VII(A) Gazette 05-Dec-2024; Kerala Electricity Duty Act 1963 (10%); IS 875 Part 3 Vb 39 m/s; IRC:46-1972; Kerala Municipality Act ss. 271–272; CBIC heading 8528. Confirm before money moves.

Annex A — Site lease agreement / mall MoU clause set

Governing law: Indian Contract Act, 1872; Kerala Municipality Act, 1994; Kerala Stamp Act, 1959; Registration Act, 1908; Electricity Act, 2003; Kerala Electricity Supply Code, 2014.

This clause set is drafted for (a) private land under an outdoor unipole in Kochi or Kozhikode, and (b) mall common-area licences for dual-sided MUPI kiosks (Lulu, Forum, HiLite, Secura/Capitol). Replace the recitals with the actual parties, survey numbers, and mall unit IDs before execution. Have a Kerala-enrolled advocate vet stamp duty and registration before signing.

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1. Parties, premises, and licence (not a tenancy of mall common area)

1.1 The **Licensor** is the owner / lawful occupant of the Premises described in Schedule A (survey number, village, taluk, corporation ward, or mall unit ID and floor).

1.2 The **Licensee** is the DOOH operator, a company incorporated under the Companies Act, 2013, with its registered office in India and GSTIN stated in Schedule A.

1.3 For mall common area, this instrument is a **leave-and-licence / display licence** and does **not** create a tenancy under the Kerala Buildings (Lease and Rent Control) Act, 1965. The Licensee acquires no interest in the land or building beyond the contractual right to install, power, operate, and maintain the Display.

1.4 For private outdoor land, if the Term exceeds eleven (11) months, the parties **shall** register this instrument under the Registration Act, 1908, at the Sub-Registrar having jurisdiction, and shall pay stamp duty under the Kerala Stamp Act, 1959, as assessed by the Sub-Registrar (working planning rate for a 1–5 year lease of immovable property: **5% of average annual rent**, plus registration fees as notified). Mall licences shorter than 11 months may be stamped as agreements; the Licensee shall not rely on non-registration to avoid payment of licence fee.

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2. Term, lock-in, and renewal

2.1 **Term:** Thirty-six (36) months from the Commencement Date, which is the later of (i) the date the Display is energised and (ii) the date the municipal advertisement permission is first issued.

2.2 **Lock-in:** Eighteen (18) months. If the Licensee exits during lock-in other than for Licensor default or Force Majeure lasting more than ninety (90) days, the Licensee shall pay liquidated damages equal to the Minimum Guarantee for the unexpired lock-in, capped at nine (9) months of Minimum Guarantee.

2.3 **Renewal:** The Licensee has a right of first refusal to renew for one further term of thirty-six (36) months at a licence fee not exceeding 110% of the fee in the final year of the then-current Term, provided the Licensee is not in material default.

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3. Commercial models (choose one in Schedule B)

3.A Fixed rent

The Licensee shall pay a monthly licence fee of the amount stated in Schedule B, plus GST at 18% (SAC 997212 / commercial rental of immovable property, or the SAC applicable to the mall's tax invoice). Fee is payable by the 7th working day of each month. Delay beyond 15 days attracts interest at 18% per annum.

3.B Revenue share

The Licensee shall pay **eighteen percent (18%) of Gross Advertising Revenue** attributable to the Premises. "Gross Advertising Revenue" means amounts actually invoiced by the Licensee for inventory on the Display, excluding GST, agency commission actually paid to unaffiliated agencies (capped at 15% of the invoice), and programmatic SSP fees actually deducted by Hivestack, Lemma, or a successor SSP. Mall reserved loop time under Clause 8 is not revenue.

3.C Hybrid (default for the pilot)

Each calendar month the Licensee shall pay the **higher of** (i) the Minimum Guarantee in Schedule B and (ii) 18% of Gross Advertising Revenue for that month. Minimum Guarantees used in the financial model:

Site	Minimum Guarantee (₹ / month, exclusive of GST)
Kochi outdoor (Edappally / Kakkanad)	80,000
Lulu Mall MUPI	40,000
Forum Mall MUPI	32,000
Kozhikode outdoor (Thondayad)	45,000
HiLite Mall MUPI	28,000
Secura / Capitol MUPI	20,000

3.4 The Licensee shall deliver a monthly revenue statement by the 10th of the following month, with proof-of-play logs. The Licensor may audit once per financial year on 15 days' notice. Overstatement variance above 5% is a material default.

4. KSEB sub-metering, billing, and surge liability

4.1 **Outdoor sites:** The Licensee shall take a dedicated LT-VII(A) three-phase connection from Kerala State Electricity Board Limited in the Licensee's name, sanctioned load 15 kW (single face) or 25 kW (dual face), and shall pay all energy charges, fixed charges, electricity duty under the Kerala Electricity Duty Act, 1963, and fuel/ FPPCA levies. The Licensor shall grant an irrevocable licence to lay cable from the KSEB metering point to the structure and shall sign Form-A / consent as owner.

4.2 **Mall sites:** Power shall be drawn from the mall LT network through a **class-1.0 sub-meter** installed at the Licensee's cost. The mall shall invoice actual kWh at the mall's KSEB LT-VII(A) energy charge plus a recovery not exceeding **20%** for HVAC common-area and transformer losses. The mall shall not levy a fixed "HVAC share" disconnected from metered consumption.

4.3 **Surge and outage:** Neither party is liable for KSEB load-shedding. The Licensee shall install Class II SPDs and ELCBs. The Licensor is liable for damage caused by defective building earthing or by water ingress from mall roof/slab leaks. The Licensee is liable for damage caused by the Display's own electrical fault, including fire originating in the cabinet, player, or PSU.

4.4 The Licensee shall maintain public liability insurance of not less than **₹2,00,00,000** per occurrence and all-risks equipment insurance for replacement value, and shall name the Licensor as an additional insured for premises liability.

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5. Installation, access, and monsoon works

5.1 Installation drawings, structural certificate, and (for outdoor) soil report and lightning design shall be delivered to the Licensor at least 21 days before mobilisation. Mall works shall occur in the night window 23:00–06:00 unless the mall operations manager agrees otherwise in writing.

5.2 The Licensor shall provide 24×7 access for a two-person maintenance crew and, for outdoor sites, crane access on not more than four occasions per year without additional fee (monsoon emergency access is unlimited).

5.3 From 1 June to 30 November each year the Licensee shall complete a pre-monsoon inspection (torque, sealant, drain holes, SPD, earthing) and issue a certificate to the Licensor within seven days.

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6. Permits and indemnity for unauthorised display

6.1 The Licensee shall obtain and keep in force: municipal advertisement permission under Sections 271–272 of the Kerala Municipality Act, 1994; KMBR building permit where the unipole is a "building" / outdoor display structure; PWD or NHAI NOC where applicable; KSEB connection; and Electrical Inspectorate scheme approval / sanction for energisation where the installation is a neon/LED advertising structure or the connected load so requires.

6.2 If any competent authority orders removal, the Licensee shall comply at its cost and this Agreement shall terminate without lock-in damages. Prepaid licence fee for the unused period shall be refunded pro rata within 21 days.

6.3 The Licensee shall indemnify the Licensor against penalties, demolition charges, and third-party claims arising from the Display, except to the extent caused by the Licensor's structural defect or failure to disclose a title / CRZ / AAI funnel restriction.

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7. Structural and weather damage (monsoon indemnity)

7.1 The Licensee warrants that the outdoor structure is designed for basic wind speed **Vb = 39 m/s** (IS 875 Part 3:2015, Kochi / Kozhikode / Kannur) with importance factor k1 not less than 1.06 for advertising structures, and that foundations are designed for the soil SBC in the geotechnical report.

7.2 **Licensee monsoon indemnity:** The Licensee shall indemnify the Licensor and the public against death, injury, and property damage caused by collapse, flying cladding, electrical flashover, or water leaking from the

Licensee's cabinet during southwest or northeast monsoon, except where collapse is caused by a third-party vehicle impact or by the Licensor's excavation within the stated influence zone of the foundation.

7.3 **Licensor monsoon indemnity:** The Licensor shall indemnify the Licensee for Display damage caused by mall roof failure, uncontrolled terrace flooding, or falling façade elements of the Licensor's building.

7.4 Neither party is liable for consequential loss of advertising revenue beyond thirty (30) days of documented outage caused by the other party's default.

8. Mall promotion inventory (reserved loop)

8.1 For mall kiosks only, the Licensor is entitled to **twelve percent (12%) of loop time** (for a 100-second loop: twelve seconds) for mall wayfinding, events, and statutory notices. This inventory is not Gross Advertising Revenue and is not set off against Minimum Guarantee.

8.2 Mall creative shall be delivered as 1080p or 4K MP4, 10 or 15 seconds, with audio optional. The Licensee may refuse creative that is obscene, political, or that violates the Cable Television Networks (Regulation) Act advertising code as applied by analogy, or that flashes above the brightness cap in Clause 9.

8.3 Emergency public-service interrupts (NDMA / district disaster management / Kerala State Disaster Management Authority monsoon alerts) may pre-empt the loop without fee for the duration of the alert.

9. Content, brightness, and traffic safety

9.1 Outdoor faces shall auto-dim: maximum 5,500 nits by day; not more than 600 nits from 22:00 to 06:00; transition not steeper than 15% per minute. No full-motion video faster than 8 frames of hard cut per 10 seconds on faces visible from a carriageway. No red/green sequential flashing that can be confused with traffic signals.

9.2 The Licensee shall not display tobacco, e-cigarettes, or any advertisement prohibited under the Cigarettes and Other Tobacco Products Act, 2003, or the Consumer Protection Act, 2019 (misleading advertisements). Liquor advertising shall comply with the Kerala Abkari Act and any municipal bye-law then in force.

10. Default, step-in, and handover

10.1 Material default includes: non-payment beyond 30 days; operation without a live advertisement licence; structural certificate older than 24 months; earthing resistance above 2 ohms left unrectified for 14 days after notice; and insolvency.

10.2 On expiry or termination the Licensee shall remove the Display within 30 days, restore the Premises (make-good of grout, paving, and mall flooring), and hand over keys. If the Licensee fails, the Licensor may remove at the Licensee's cost after 7 days' notice.

11. Dispute resolution

11.1 This Agreement is governed by the laws of India. Courts at **Ernakulam** (for Kochi sites), **Kozhikode** (for Kozhikode sites), or **Thalassery / Kannur** (for Kannur sites) have exclusive jurisdiction.

11.2 Disputes shall first be referred to mediation under the Mediation Act, 2023. Failing settlement in 30 days, arbitration by a sole arbitrator under the Arbitration and Conciliation Act, 1996, seat **Kochi**, language English.

Schedule A — Premises (complete before execution)

- Survey No. / Re-Sy No. / mall unit ID: as identified in the sanctioned drawing.
- Extent of licensed footprint: outdoor 6.0 m × 6.0 m exclusive of guy/crane swing; indoor 1.2 m × 1.2 m plus 0.5 m service clearance.
- Title document / mall lease: Licensor represents it has the right to grant this licence and that the Premises are not in a CRZ-I prohibited area and, for Kochi outdoor, that AAI NOCAS has been checked for the proposed height.

Schedule B — Commercials

- Model: Hybrid / Fixed / Revenue share (strike out unused).
- Minimum Guarantee and revenue-share percentage as Clause 3.
- Security deposit: three (3) months of Minimum Guarantee, interest-free, refundable within 30 days of handover subject to make-good.
- GST invoices: Licensor to issue tax invoice; Licensee to issue advertising tax invoices to brands.

Annex B — Structural stability certificate (IS 875 / KMBR)

To be issued on the letterhead of a **Chartered Structural Engineer** (M.Tech Structural / equivalent) registered with the concerned Municipal Corporation / Local Self Government Institution, with valid Institution of Engineers (India) membership. Attach STAAD/ETABS or equivalent printouts, geotechnical report, and fabrication drawings.

Certificate No.: SSE/DOOH/{{YEAR}}/{{NNN}}
Date: {{DATE}}
Project: Dual-face P3.91 outdoor LED unipole, 8.0 m × 4.0 m per face, mounting height to underside of panel 9.0–12.0 m.
Site: {{SURVEY_NO}}, {{VILLAGE}}, {{CORPORATION_WARD}}, {{CITY}} (Kochi / Kozhikode).
Owner / Operator: {{COMPANY_NAME}}, CIN {{CIN}}, GSTIN {{GSTIN}}.

1. Codes applied

Code	Use
IS 875 (Part 3):2015	Wind loads. Basic wind speed **Vb = 39 m/s** for Kochi, Kozhikode, and Kannur (50-year return, 10 m, Category 2).
IS 875 (Part 1 & 2)	Dead load of cabinets, steel, walkway; imposed load 0.75 kN/m² on maintenance platform.
IS 1893 (Part 1): 2016	Seismic — Kerala Zone III, importance factor 1.2 for advertising structures adjacent to public roads.
IS 800:2007	Steel design (limit state).
IS 456:2000	RCC foundation.
IS 2062:2011	Structural steel grade E250 / E350 as specified on drawings.
IS/IEC 62305	Lightning protection Level III minimum.
KMBR 2019	Outdoor display structure treated as a “building” / advertising sign under Rule 2 definitions.

2. Design wind (worked example for a 12 m hub height, Terrain Category 2)

$V_z = V_b \times k_1 \times k_2 \times k_3 \times k_4$
$V_z = 39 \times 1.06 \times 1.00 \times 1.00 \times 1.00 =$ **41.34 m/s**
Design pressure $p_z = 0.6 \times V_z^2 = 0.6 \times 1{,}708.9956 =$ **1.025 kN/m²**
Force-coefficient C_f for a sharp-edged rectangular billboard: **1.20** (IS 875 Part 3, Table 23 / equivalent).
Area per face (including 50 mm cabinet frame): 8.10 m × 4.10 m = 33.21 m².
Along-wind force per face $F_w = p_z \times C_f \times A = 1.025 \times 1.20 \times 33.21 =$ **40.82 kN**.
Dual-face: design for **vector combination** of both faces with a 15° yaw case and a 0° case; torsion from unequal dimming/soiling included as 10% eccentricity.

The undersigned certifies that member stresses and foundation bearing / uplift remain within allowable limits for the above loading, including a monsoon residual-life case with 20% section-loss on unprotected welds (the specification nonetheless requires a full coastal paint system so that this case is a robustness check, not the intended corrosion state).

3. Geotechnical

A borehole investigation of not less than **two boreholes to 8 m or refusal**, with SPT at 1.5 m intervals, has been reviewed. Allowable net SBC used in design: **{{SBC}} kN/m²** at founding level **{{DEPTH}} m**. For coastal / backwater-influenced sites (Edappally canal belt, Kozhikode coastal plain), the foundation shall be **RCC raft or bored piles** and shall not be a shallow isolated footing if the water table is within 2.0 m of founding level.

4. Lightning and earthing (structure)

Air-termination: 10 mm Cu rod or 25 × 3 mm GI strip on the highest metallic point, down-conductor 25 × 3 mm GI or 35 mm² Cu, test link at 1.5 m AGL, earth electrode pit with 600 × 600 × 3 mm Cu plate or 3 m copper-bonded rod, **resistance ≤ 2 ohms** in dry season, confirmed by fall-of-potential test. Bond all cabinets, walkway, and LT panel to the same earth grid. Isolate data/power SPDs to the same bar.

5. Declaration

I, {{ENGINEER_NAME}}, Chartered Engineer (India) M-{{MEMBERSHIP}}, having independently checked the design and (strike out one) visited / not yet visited the site, certify that the structure as shown on drawings {{DWG_NOS}} is safe for the intended advertising use, will not endanger the public or adjoining property under the design monsoon wind, and is fit for grant of building permit / advertisement permission.

This certificate is valid for **twenty-four (24) months** from the date above or until a material alteration, whichever is earlier. A monsoon re-inspection certificate shall be issued by 15 May each year.

Signature / seal of Structural Engineer
Name, address, mobile, email, corporation registration number.

Countersignature of Licensee’s authorised signatory
Accepting that operation at brightness or cabinet weight above the certified envelope voids this certificate.

Annex C — Factory QC and Cochin Port import checklist

Use this checklist at the Shenzhen / Guangzhou factory **before** containers are sealed, and again at ICTT Vallarpadam / CFS before the goods leave Cochin Port.

A. Factory acceptance tests (FAT)

Test	Method	Pass criterion
Pixel / module map	Novastar receiving-card scan	Zero dead pixels on a 32 sq.m face; ≤ 3 field-replaceable modules with cosmetic blemish
Brightness	Calibrated luminance meter, 10-point grid, 5,500 nits mode	Average ≥ 5,500 nits; corner-to-centre uniformity ≥ 90%
IP rating	IEC 60529	Cabinet IP65 minimum (IP66 preferred). Power/data connectors IP67. Drain holes proven open.
Water	Spray + standing water on top face 1 hour	No ingress to PSU or receiving card; silica-gel packs still dry
Burn-in	72 hours continuous, 30 min 100% white / 30 min video loop	No thermal trip; cabinet back < 60 °C ambient 35 °C
Salt spray	ISO 9227 / ASTM B117, 96 hours on spare module and cabinet cut-sample	No red rust on die-cast Al; coating adhesion ASTM D3359 4B or better
Conformal coat	IPC-CC-830	Silicone (SR) coat on outdoor PCBs, not acrylic, coverage to IPC-A-610 Class 2
Power	Mean Well HLG / equivalent, 90–305 VAC	PSU labelled, serialised, 7-year OEM warranty letter
Spare kit	Pack-out list	5% extra modules per face, 2 PSU per face, 1 receiving card per face

Photograph every serial plate. Seal the container with a numbered bolt seal and record it on the packing list.

B. Documents that must travel with the cargo

1. Commercial invoice with model, pixel pitch, nits, IP rating, country of origin **China**, and **CTH 8528 52 00** (or 8528 59 00 if customs insists — see blueprint Module 2).
2. Packing list with cabinet IDs and net/gross weight.
3. Bill of lading (FCL, port of loading Shekou / Yantian / Nansha; port of discharge **INCOK — Cochin / ICTT Vallarpadam**).
4. Insurance (ICC-A, 110% of CIF).
5. Certificate of Origin (Form F / general, as applicable).
6. OEM test reports (IP, brightness, salt spray) and CE / RoHS declarations.
7. Importer IEC, GSTIN, AD code, and CHA authorisation.
8. Purchase order and signed technical specification (this checklist attached).

C. Cochin Port / ICEGATE sequence

1. Shipping line files IGM. Track on ICEGATE: https://www.icegate.gov.in
2. CHA files Bill of Entry (self-assessment) with CTH, BCD, AIDC, SWS, IGST.
3. Pay duty via ICEGATE e-payment. IGST is claimed as ITC in the next GSTR-3B.
4. If selected for examination, insist on a dry, covered exam bay — monsoon rain on open cabinets voids IP warranty.
5. DP World ICTT gate-out: https://www.dpworld.com/en/ports-terminals/india/ictt/services
   Cochin Port Authority: https://www.cochinport.gov.in
6. Move on a covered trailer to the Kochi warehouse. Do not store cabinets outdoors in June–November even if IP65.

D. Site incoming inspection

Megger incoming cables, confirm earth ≤ 2 Ω, 72-hour on-site burn-in before declaring commercial operations, and keep FAT serials matched to the installed map.

End of report. Regenerated with `python3 scripts/build_pdf_report.py`. Figures in Module 4 and the executive summary are live outputs of `dooh_kerala_calculator.py`.